

Multiple Choice (10 marks)

1. Consider the Bayesian network given below. How many independent parameters are needed for this Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 2
 - (b) 4
 - (c) 8
 - (d) 16
2. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
3. For polynomial regression, which one of these structural assumptions is the one that most affects the trade-off between underfitting and overfitting:
 - (a) The polynomial degree
 - (b) Whether we learn the weights by matrix inversion or gradient descent
 - (c) The assumed variance of the Gaussian noise
 - (d) The use of a constant-term unit input
4. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
5. Which among the following prevents overfitting when we perform bagging?
 - (a) The use of sampling with replacement as the sampling technique
 - (b) The use of weak classifiers
 - (c) The use of classification algorithms which are not prone to overfitting
 - (d) The practice of validation performed on every classifier trained

True / False (10 marks)

Answer True or False

6. True or False: Grid search cannot be applied to non-differentiable functions, limiting its use in various optimization scenarios.
7. True or False: All methods, including PCA, can be categorized as supervised learning approaches in machine learning.
8. True or False: As of 2020, certain models achieve over 98% accuracy on the CIFAR-10 dataset, demonstrating significant advancements in image classification.
9. True or False: The model obtained by applying linear regression may differ during forward stage wise selection compared to the final subset model.
10. True or False: Residual connections are commonly utilized in both ResNets and Transformers to facilitate training of deep networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how you would modify the EM algorithm to compute MAP estimates instead of maximum likelihood estimates, focusing on the necessary changes to the maximization step.
12. Discuss the implications of applying regularization in linear regression, specifically addressing how certain penalties can lead to some coefficients being zeroed out.

End of Paper